

# Feasibility of an AI-Assisted Judicial Research Platform for the Nepalese Legal Domain

## A Proposed Judicial AI Framework for Nepal

*Concept and Technical Paper*

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### 1. Abstract

Nepal is putting more of its laws, court judgments and case records online, but this information is still scattered across different websites, formats and languages. Finding a document is only the first step in legal research. A trustworthy system also must answer harder questions: Is this source official? Was this law in force on the date that matters? Does this judgment carry real legal authority? And is every AI-generated statement backed up by the source it cites?

This paper asks whether an AI-assisted Judicial AI platform is realistic for Nepal. The proposed design combines verified legal data, legal metadata, a mix of keyword and meaning-based search (“hybrid retrieval”), an “Authority-Aware” version of Retrieval-Augmented Generation (RAG), large language models (LLMs), citation checking, security controls, and human oversight. The system is meant to support legal research and decisions — not to act as an independent judge.

The paper reviews international judicial-AI practice, looks closely at recent AI-related developments inside Nepal's own judiciary, identifies specific gaps in Nepal's legal data and institutions, lays out a practical system design, and evaluates whether the idea is feasible technically, operationally, legally and ethically. The main finding: a small, carefully governed Judicial AI prototype is realistic for Nepal — if it starts with verified public legal sources, clear rules about legal authority and law versions, measurable testing, and strong human control throughout.

*Keywords: Judicial AI; Nepal Judiciary; Legal RAG; Large Language Models; Legal Information Retrieval; Authority-Aware Retrieval; Legal Metadata; Human Oversight.*

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## 2. Introduction and Background

AI is increasingly used to search, sort, summarize and explain large sets of documents. This is useful in law because legal research depends on huge volumes of Acts, regulations, judgments, amendments and procedural rules. But law is not like ordinary document search. A piece of text can closely match a user's question and still be legally wrong — because it is outdated, from the wrong jurisdiction, or missing an important exception. For this reason, a Judicial AI system needs to do more than a general-purpose chatbot.

Retrieval-Augmented Generation (RAG) is a good starting point, because it lets an LLM read real documents instead of relying only on what it memorized during training [1]. Research on legal-domain language models also shows that training or adjusting a model specifically on legal text improves its performance on legal tasks [2]. But neither RAG nor a legal LLM can, on its own, decide what counts as legal authority, track amendment history, know whether a precedent still applies, or know what information is confidential. These checks must be built around the model, not left inside it.

### 2.1 Nepal's Judiciary and Legal System

Nepal has a constitutional judiciary led by the Supreme Court, followed by High Courts and District Courts. Legal information is also published by other public bodies, most importantly the Nepal Law Commission, which maintains the official repository of the Constitution, Acts, rules and other legal texts [3]. The Supreme Court provides judgments, case status and other services through its own digital systems, and its Fifth Five-Year Strategic Plan continues a broader push towards modernization and better use of information systems [4].

In July 2026, the Supreme Court's full court took this further by adopting a new Judiciary Information and Communication Technology Policy. For the first time, this policy formally allows AI to be used for legal research, case management and administrative work — while explicitly keeping AI out of judgments, writ petitions and any other decision that requires judicial discretion [16]. This recent step is directly relevant to the framework proposed later in this paper and is discussed further in Section 4.4.

Nepal's legal domain is also multilingual and structurally complex. Nepali is the main language of most Acts and judgments, but lawyers, researchers and citizens often search in English or in a mix of both. The exact legal meaning of a text can depend on its section, subsection, proviso, explanation, schedule, amendment date, or a later judgment. A useful AI system must preserve these relationships rather than treating every document as plain, undifferentiated text.

### 2.2 Current Legal Information Challenges

Nepal already has valuable legal information online, but legal research still takes a lot of manual effort. The main challenges are:

- Legal material is spread across many different websites, databases, PDFs and scanned documents.
- Search works well mainly when the user already knows an exact case number, section, party name or legal term.
- Some older documents have OCR errors, font issues or broken Unicode text.
- It is hard to follow amendments and older versions of a law in one continuous view.
- A relevant-looking judgment may not actually be binding — it may have been overruled, distinguished, or decided under an earlier legal framework.

- People ask questions in Nepali, English, or a mix of both, while the source document may exist in only one language.
- No existing system brings the rule, its exceptions, its amendments, related precedents and the exact cited paragraph together in one place.

## 2.3 Why AI-Assisted Legal Analysis Matters

The goal is not to replace judges or lawyers. It is to reduce the time spent finding documents, comparing laws and doing repetitive research — while keeping final legal responsibility firmly with people. A well-built system could support legal search, case summarization, citation checking, amendment tracking, multilingual access and research preparation. International guidance from UNESCO and the Council of Europe backs this same human-led, rights-respecting approach to AI in justice systems [5][6].

## 3. Problem Statement and Research Objectives

The core problem is that a general-purpose LLM can write fluent, legal-sounding text without knowing whether its source is authoritative, still in force, or applicable to the question. Ordinary RAG improves this by grounding answers in real documents, but it usually ranks text mainly by how closely it matches the question. In judicial work, matching alone is not enough — the system also needs to understand legal sources, court hierarchy, jurisdiction, timing, how a precedent has been treated since, and whether the evidence genuinely supports the claim.

The research question is: How feasible is an AI-assisted Judicial AI platform for Nepal that can retrieve and analyze legal information while preserving legal authority, correct timing, citation support, security and human accountability?

The objectives of this paper are to:

1. Describe Nepal's legal-information problem and the opportunity for AI-assisted research.
2. Review the key technical foundations and international Judicial AI initiatives.
3. Identify Nepal's specific research and implementation gap, including recent local developments.
4. Propose a practical Judicial AI concept, scope and system architecture.
5. Assess technical, data, operational, legal and ethical feasibility.
6. Define the implementation of roadmap, governance model and safety constraints.
7. Identify the major challenges, limitations and future research needs.

## 4. Literature Review and Technical Foundations

### 4.1 Artificial Intelligence in Legal Systems

AI is already used in legal work for document classification, legal search, translation, summarization, prediction, drafting support and administrative tasks. Judicial systems that have successfully adopted AI usually start with narrow, well-defined tasks rather than letting AI make decisions on its own. Brazil's Supreme Federal Court uses a system called VICTOR to help classify cases linked to general repercussion appeals [7]. Singapore's Small Claims Tribunals use generative AI for translation and case-document summarization, including support for people

representing themselves without a lawyer [8]. India's Supreme Court runs several digital services, including an AI assistant called SuSahayak, while making clear that the original court record remains the final authority [9].

These examples share a pattern: AI works better, and is trusted more, when the task is narrow, the data source is controlled, the output can be checked, and a person stays responsible for the outcome. This pattern applies directly to Nepal.

## 4.2 LLMs, RAG and Legal Information Retrieval

LLMs are good at generating language, summarizing text and reasoning in general — but they can also invent facts, rely on outdated information, or produce citations that look correct but are not. RAG helps by pulling in real evidence before the model writes its answer [1]. In legal work, though, retrieval needs to do more than find similar-sounding text. Exact section numbers, case names, dates and legal phrases matter a great deal, so a hybrid approach — combining keyword search with meaning-based (vector) search — works better than either method alone.

Legal-focused NLP research supports this. LEGAL-BERT showed that training or adapting a model specifically on legal text improves results on legal tasks [2]. Benchmark competitions such as COLIEE have shown that finding prior cases and reasoning about legal entailment are genuinely hard problems that need their own dedicated tests, separate from general-purpose language benchmarks [12]. More recent research on legal RAG systems shows that retrieval quality, how much relevant evidence is found, and how well citations are supported often matter more to final answer quality than the raw capability of the language model itself. This supports a design where retrieval, authority-checking and citation verification are each tested separately, instead of being folded into one overall AI score.

## 4.3 International Judicial AI Initiatives

No single country's model should be copied directly into Nepal, but each offers a lesson. Brazil shows the value of narrow, high-volume case classification. Singapore shows how generative AI can be used carefully for translation and summarization. India shows how multilingual legal information and public digital services can work together. The Council of Europe's CEPEJ Ethical Charter states that AI can improve justice services, but only while respecting fundamental rights, non-discrimination, quality, transparency and user control [6]. UNESCO's 2025 Guidelines for the Use of AI Systems in Courts and Tribunals lay out a similar global framework built around human oversight, local governance, transparency and responsible use [5].

## 4.4 Nepal's Emerging Judicial AI Policy and Practice

Nepal's own judiciary and government have already begun moving on this issue, and these recent, local developments matter a great deal for how any Judicial AI platform should be designed.

At the national level, the Government of Nepal approved its first National Artificial Intelligence Policy in August 2025. It sets out a broad framework for AI ethics, infrastructure, skills and governance across all sectors of the economy [16]. However, this national policy does not contain specific rules for using AI inside courts — that gap has been left for the judiciary to fill on its own.

The judiciary has started to fill it. In July 2026, the Supreme Court's full court adopted the Judiciary Information and Communication Technology Policy, 2026. This policy formally allows AI to be used for legal research, case management, and administrative and academic work, but it explicitly prohibits AI from writing judgments,

deciding writ petitions, or performing any task that requires judicial discretion [17]. This is a real, independent confirmation — arrived at by Nepal's own judiciary, not by this paper — of the same assistive, human-controlled approach proposed in Section 5.

Nepal's judiciary has also engaged directly with international guidance. In late 2025, judges, senior advocates, government attorneys, and representatives of the Supreme Court, the Nepal Bar Association and the National Judicial Academy met in Kathmandu for the country's first national consultation on UNESCO's draft Guidelines for the Use of AI Systems in Courts and Tribunals [18]. A related UNESCO survey found that most judicial staff were already familiar with AI tools, and a large share reported already using general-purpose tools informally in their day-to-day work [18]. In other words, some form of AI use is already happening inside Nepal's legal system, whether a formal, governed platform exists to support it safely.

The case for a structured system is reinforced by Nepal's case backlog problem. Close to 40 percent of cases before Nepal's judiciary remain unresolved at any given time, and weak case management is one of the most-cited reasons for delay [19]. Commentators have suggested that automated case classification, workload prediction and AI-assisted scheduling could meaningfully help reduce this backlog [19], all of which line up with the assistive, non-decisional functions proposed in Section 5 of this paper.

At the same time, Nepali commentary on this topic raises real concerns that any proposed framework needs to take seriously. Legal writers have pointed out that Nepal's courts already use automated systems — for example, an automated case-listing system now used across the High Courts — with little public discussion of how these systems are supervised or who is accountable for their results [20]. Others note that lawyers, government attorneys, bench assistants and law students across Nepal are already using generative AI informally to help draft petitions and pleadings, with no regulatory body currently reviewing this practice [21]. A further concern is that most AI systems are trained mainly on dominant, high-resource languages and may lose important nuance when applied to Nepal's multilingual courtroom environment, where cases are argued across many languages, accents and forms of expression [20].

Taken together, this recent activity — national policy, judiciary policy, pilot consultations and public debate — shows that the question this paper studies is no longer hypothetical. Nepal's judiciary has already decided, in principle, that AI belongs in legal research and case administration but not in judicial decision-making. What is still missing is the detailed technical and governance design needed to make that principle work safely in practice: how to verify sources, track legal authority and timing, check citations, and keep a complete audit trail. That detailed design is the specific contribution of this paper.

Existing research on legal RAG, legal-domain language models, court technology and AI governance has, until now, mostly been studied separately. This paper's contribution is to bring all of these pieces together — verified legal ingestion, bilingual search, legal metadata, authority-aware retrieval, temporal validity, precedent relationships, grounded LLM analysis, claim-level citation verification, security, governance and human review — into one framework built specifically around Nepal's legal system, institutions and current policy direction.

## 5. Proposed Judicial AI Concept and Scope

The proposed Judicial AI platform is an assistive legal-information and research system. It connects official Nepalese legal sources, organizes them into a structured knowledge base, retrieves evidence that is both relevant and legally applicable, and uses an LLM to explain that evidence in plain language. The LLM is one component of the system — it is not the authority of the system.

The initial scope should focus on public legal information and professional legal research. Suitable functions include:

- searching Acts, regulations and judgments in Nepali and English;
- finding related precedents and legal provisions;
- showing amendments and historical versions of a law;
- summarizing judgments and comparing authorities;
- checking whether a citation exists and supports a claim;
- creating a structured evidence package for legal research;
- providing plain-language public legal information, with its limits clearly stated.

The system should not independently decide guilt or liability, write final judgments, impose sentences, judge witness credibility, or exercise judicial discretion. Any higher-risk function should require explicit institutional policy and human review — consistent with the Supreme Court's own 2026 ICT policy described in Section 4.4 [17].

## 6. Proposed Architecture and System Design

The architecture is built in clear layers, so that legal rules and governance checks cannot be skipped just because a model's response sounds fluent and confident. Figure 1 shows the proposed design.

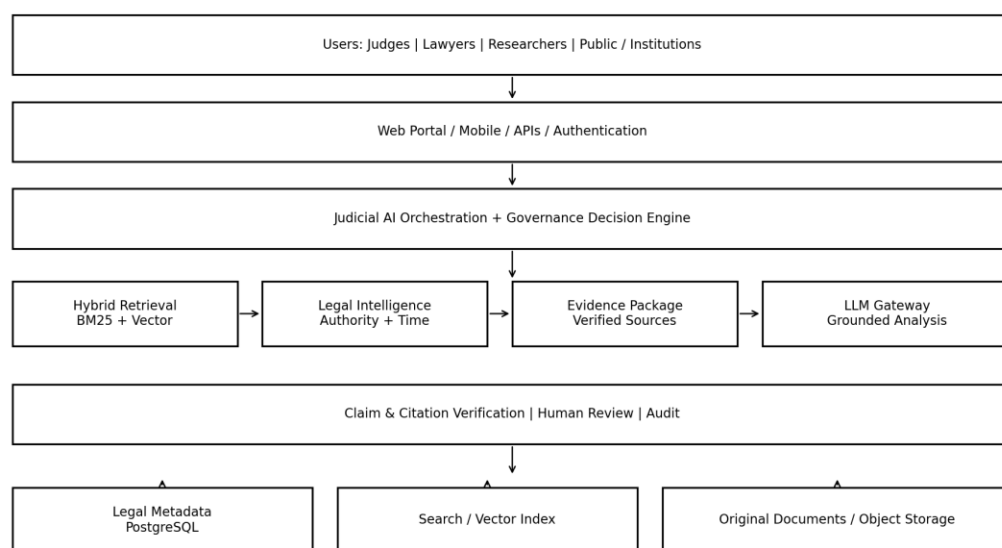


Figure 1. Proposed high-level Judicial AI architecture for Nepal.

### 6.1 Data Sources and Digitization

Priority data should come from authoritative public sources: the Nepal Law Commission, Supreme Court repositories, official judgments, the Constitution, Acts, regulations, amendments, and approved judicial publications [3][4]. The original file must always be kept unchanged and available. Every document should get a permanent internal ID, a source reference, a file hash, a verification status and version information.

Scanned documents need OCR, but OCR text should never automatically be treated as authoritative. Critical fields — case numbers, court, date, section number, amendment date — need stronger, separate validation. Low-quality pages should be flagged for a human to check.

## 6.2 Legal Document Processing

Documents should be processed according to their structure, not treated as flat text. Acts should preserve their internal Chapter, Section, Subsection, Proviso, Explanation and Schedule structure. Judgments should preserve page and paragraph references and, where possible, distinct sections such as facts, issues, arguments, reasoning, holding and order. This structure-aware approach reduces the risk of retrieving one sentence while losing the exception or context around it.

The ingestion flow is: source registration → file capture → hashing → OCR / Unicode normalization → structure detection → metadata extraction → citation extraction → human validation → legal chunking → indexing → embedding → controlled publication.

## 6.3 Knowledge and Metadata Layer

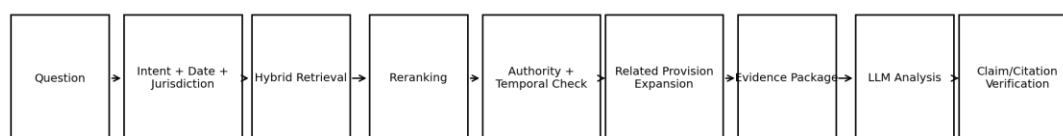
The metadata layer is what gives the documents legal meaning. It should record document type, title, court or issuing authority, jurisdiction, case number, decision date, publication date, effective date, language, legal status, source, verification status, cited laws, cited judgments and document structure. It should also track relationships such as amendment-of, cites, followed-by, overruled-by, related-provision and historical-version-of.

A relational database such as PostgreSQL can act as the single source of truth. Original files can live in object storage. Search indexes and vector indexes are derived from copies that can always be rebuilt from the verified source data.

## 6.4 Retrieval and RAG Layer

The retrieval layer should combine exact and meaning-based search. Keyword retrieval (such as BM25) is useful for case numbers, legal phrases and exact provisions. Vector retrieval is useful when a user describes an issue in their own words. Results from both should then be combined and reranked.

The proposed Authority-Aware RAG adds a legal check before evidence ever reaches the LLM. It asks two separate questions: Is this text relevant? And is it legally applicable? The second question weighs jurisdiction, court hierarchy, precedent status, effective date, amendment history, source verification and any conflicts between sources.



If evidence is weak, outdated, conflicting, or unverified: show limitation, request review, or safely refuse.

Figure 2. Proposed Authority-Aware Judicial RAG flow.



When a provision is retrieved, the system should also look for its definitions, exceptions, provisos, schedules, cross-references, related regulations, amendments and relevant judicial interpretations. Repealed law should stay available for historical questions but should never be shown as current law by default.

## 6.5 LLM Analysis Layer

The LLM receives a small, structured evidence package — not open access to the whole database. The package should contain the question, jurisdiction, relevant date, selected provisions, judgments, authority labels, conflicts and known gaps. The model's job is then to summarize or explain that material, not to search for it independently.

A separate verification step should check every material claim. Citation verification should confirm at least four things: the source exists, the cited text is represented accurately, the source applies to the claim, and the source was valid for the relevant legal period. If the evidence is incomplete, the system should say so — or safely decline to answer — rather than guessing.

## 6.6 User Interface and APIs

The interface should show the answer and its supporting evidence side by side. A user should be able to open the exact source page or paragraph, see whether a source is current or historical, and know when a translation is unofficial. Professional and public users may see different levels of detail, but both should rely on the same verified evidence underneath.

The system should be built so that the underlying AI model can be replaced or upgraded without redesigning the whole platform — legal knowledge should live outside the model, in controlled, verifiable repositories, connected through versioned APIs. A suggested technical stack for a first prototype is given in Appendix A.

## 6.7 Security and Governance

Security and governance must be designed in from the beginning, not added later. Users should be separated into public, professional, judicial, administrative and audit roles. Sensitive records need strong authentication, least-privilege access, encryption, audit logs and controlled retention. Prompts retrieved evidence and generated output can themselves contain confidential information and must be protected accordingly — the OWASP Top 10 for Large Language Model Applications is a useful reference for the specific security risks, such as prompt injection and data leakage, that LLM-based systems like this one face [13].

The governance service — not the LLM — should decide whether a request is allowed, whether an external AI provider may be used, whether a human must review the answer, and whether the answer is released, blocked, or sent for further review. NIST's AI Risk Management Framework, and its companion profile for generative AI, offer a useful general model for governing, mapping, measuring and managing these risks [10][11], while UNESCO and CEPEJ provide justice-specific principles [5][6].

# 7. Feasibility Analysis

Feasibility needs to be judged across technology, data, institutions, law and ethics. A system can be technically possible and still fail if official data is incomplete, governance is weak, or users simply do not trust it.

## 7.1 Technical Feasibility

The main technical building blocks already exist: OCR, multilingual embeddings, search engines, vector databases, rerankers, LLMs, API frameworks, identity systems and monitoring tools. A prototype can be built using mature, widely available open-source and commercial technologies (see Appendix A for a suggested stack). The hard part is not basic software availability — it is reliable legal integration, evaluation and governance. More complex technologies, such as dedicated graph databases, specialized vector clusters or fine-tuned local LLMs, should only be introduced once testing shows a real need for them.

## 7.2 Data Feasibility

Nepal has enough public legal material to build a limited research prototype. The Constitution, selected Acts, amendments and a controlled set of Supreme Court judgments can form the first dataset. The bigger challenge is data quality: historical scans, inconsistent formatting, missing metadata and unclear amendment relationships will all require human validation. Data feasibility is therefore moderate-to-high for a bounded prototype, but lower for immediate, national-scale coverage.

## 7.3 Operational and Institutional Feasibility

Operational feasibility depends on who owns the project. A successful pilot needs legal experts, court IT staff, data engineers, AI engineers, security specialists and evaluators working together. Legal experts must be involved in metadata design, benchmark creation, authority rules and error analysis — not brought in only at the final review stage. Integration with existing court and law repositories should happen through controlled APIs, rather than by building yet another isolated system.

## 7.4 Legal and Ethical Feasibility

An assistive Judicial AI system is far more feasible than an autonomous decision-making one. Legal and ethical feasibility improves when the system uses public or authorized data, keeps a complete audit trail, explains its sources, protects privacy, separates information from judicial decision-making, and requires human responsibility for any higher-risk use. Nepal's Privacy Act, 2075 [14] and Electronic Transactions Act, 2063 [15] are directly relevant to how the system handles data and electronic records, and the judiciary's own 2026 ICT policy adds a further, court-specific layer of rules [17]. International guidance also strongly supports human oversight and transparent use [5][6].

| Area            | Current Feasibility                | Main Strength                                 | Main Risk                                                 |
|-----------------|------------------------------------|-----------------------------------------------|-----------------------------------------------------------|
| Technical       | High for a prototype               | Search, retrieval and API tools are mature    | Legal reliability and multilingual accuracy               |
| Data            | Moderate                           | Official public sources already exist         | OCR errors, missing metadata, incomplete versions         |
| Operational     | Moderate                           | Judiciary already runs digital systems        | Ownership, staffing, and getting staff to actually use it |
| Legal / Ethical | Moderate-to-high for assistive use | Human-led research support is easy to control | Privacy, bias, over-reliance and accountability           |

*Table 1. Summary feasibility assessment for a bounded Judicial AI prototype.*

## 8. Implementation Roadmap, Governance and Safety Constraints

The safest path is a gradual one. The project should prove the quality of its data and retrieval before adding more complex generative features.

1. Governance and scope: define permitted uses, prohibited uses, data owners, user roles and success criteria.
2. Verified legal-data foundation: ingest the Constitution, selected Acts, amendments and 50–100 verified Supreme Court judgments.
3. Baseline search: build exact, filtered and keyword search, and set measurable retrieval targets.
4. Hybrid retrieval: add multilingual vector search, result fusion and reranking.
5. Legal intelligence: add jurisdiction, court hierarchy, temporal validity, precedent status and related-provision rules.
6. Grounded generation: give the LLM a controlled evidence package and require citations for every claim.
7. Verification and safety: add claim checking, citation validation, adversarial testing, audit logging and human review.
8. Controlled pilot: test with a small group of judicial researchers or legal professionals before wider rollout.
9. Scale only after evidence: expand datasets, integrations and infrastructure only after independent evaluation.

Safety constraints should be explicit and non-negotiable. The system should never invent citations, never present unverified OCR text as authoritative, never silently apply repealed law to a current question, never expose confidential material to unapproved model providers, and never make judicial decisions on its own. Where evidence is weak or conflicting, the correct behavior is to show that uncertainty or ask for human review — not to guess.

## 9. Challenges and Limitations

This framework has real limitations. First, this paper presents a concept and an architecture — not a finished, production-ready system. The proposed scores, rules, thresholds and technologies all still need to be tested in a working prototype. Second, Nepal does not yet have a large public benchmark for bilingual legal retrieval, authority classification, historical-law questions or claim-level citation checking. Third, legal data quality may be inconsistent, especially for older documents.

Legal authority is also hard to automate fully. A judgment may look relevant but still be distinguishable on its facts or procedure. How a precedent is treated can depend on bench composition, later cases, and professional interpretation. Translation can quietly shift legal nuance. Even a correct source can be applied to the wrong situation by a language model. For all these reasons, this architecture can reduce risk — it cannot remove the need for human legal judgment.

Other challenges include institutional resistance, the cost of cleaning historical data, cybersecurity, privacy, dependence on outside vendors, model changes over time, uneven performance in Nepali, and automation bias

— the tendency to trust a polished AI answer more than it deserves. The interface and any staff training programme need to actively encourage source-checking, not passive trust.

## 10. Discussion

The central design choice in this paper is to treat Judicial AI as a complete legal-information system, not just an LLM application. That changes where trust comes from. Trust should not come from the model sounding confident — it should come from verified sources, legal metadata, retrieval quality, authority checks, version control, citation verification, audit trails and human responsibility.

This approach adds some complexity and a little extra response time. But it also makes errors much easier to trace. If an answer is wrong, the team can ask exactly where the problem came from: OCR, metadata, retrieval, authority classification, choosing the wrong version of a law, generation, or verification. That is far more useful than treating every failure as a vague “hallucination.”

The architecture also fits Nepal's real situation. Nepal does not need to train its own national foundation model before it can start Judicial AI research. A small, well-governed RAG prototype can use existing models while keeping legal knowledge outside the model, in controlled repositories. Fine-tuning or local models can be considered later, once benchmark results show a clear benefit.

International experience points the same way. Brazil, Singapore and India show that courts gain real value from classification, translation, summarization and controlled digital assistance [7][8][9]. Nepal's own judiciary has already reached a similar conclusion in practice: its 2026 ICT policy permits AI for research and administration while keeping it out of judgments [17]. The framework proposed here builds directly on that same direction, and adds the specific technical components — legal authority, temporal validity and citation verification — needed to make it work safely.

## 11. Conclusion and Future Work

This paper finds that AI-assisted legal analysis is feasible for Nepal at the level of a controlled research prototype, and, with institutional support, a later judicial pilot. Nepal already has important digital legal sources, a continuing judicial modernization effort, and — as of 2026 — a formal judiciary policy that welcomes AI for research and administration. Modern search, RAG and LLM technologies are mature enough to build a genuinely useful prototype. The main barriers are legal-data quality, authority and version modelling, multilingual evaluation, governance, security and sustained expert involvement.

The proposed Judicial AI platform therefore follows a layered design: verified legal data → structured metadata → hybrid retrieval → authority and temporal checks → evidence package → grounded LLM analysis → claim and citation verification → human review and audit. The system is meant to assist legal research, not to replace judicial discretion.

Future work should build a Nepal-specific benchmark and prototype. The benchmark should include paired Nepali English questions, current and historical law, related provisions, conflicting judgments, fake citations, unanswerable questions and adversarial document changes. The prototype should compare keyword, vector, hybrid, reranked and authority-aware retrieval against each other. Later research can look at precedent graphs, domain-adapted models, privacy-preserving deployment, public-facing legal information, institutional adoption,

and eventually national-scale integration — building on the direction the Supreme Court has already set with its 2026 ICT policy.

The central principle is simple: legal trust should come from verified evidence and an accountable process — not from confident-sounding AI writing.

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## Appendix A: Suggested Technical Implementation Stack

These are illustrative technology choices for a first prototype, moved here from the main body so that the core argument stays readable for non-technical audiences. Nothing below is a fixed requirement — the architecture in Section 6 is designed to work with any comparable set of tools, and choices should be revisited once prototyping begins.

- Web interface: Angular, or a comparable modern web framework.
- Enterprise APIs, identity and workflow: ASP.NET Core (or an equivalent enterprise framework).
- AI processing services: Python with FastAPI.
- Source of truth / structured data: PostgreSQL, using the pgvector extension for embeddings.
- Full-text and keyword search: OpenSearch, or a comparable search engine.
- Original file storage: object storage (e.g., an S3-compatible service).
- Deployment: containerized services (Docker); Kubernetes, dedicated graph databases, or specialized vector-database clusters only once testing shows a genuine need.
- Integration: versioned REST APIs connecting ingestion, search, research, verification and review services, so that any single component — including the LLM provider — can be replaced without redesigning the whole platform.

## Appendix B: Glossary of Terms

| Term                                        | Plain-language meaning                                                                                                                                                                                              |
|---------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>AI (Artificial Intelligence)</b>         | Computer systems are designed to perform tasks that normally require human intelligence, such as understanding language or recognizing patterns.                                                                    |
| <b>LLM (Large Language Model)</b>           | An AI model trained on large amounts of text that can generate, summarize and explain language, such as ChatGPT or similar systems.                                                                                 |
| <b>RAG (Retrieval-Augmented Generation)</b> | A method where an AI model is given real, retrieved documents to read before it answers, instead of relying only on what it learned during training.                                                                |
| <b>Authority-Aware RAG</b>                  | The version of RAG proposed in this paper, which checks not just whether retrieved text is relevant, but whether it is legally applicable — considering jurisdiction, court hierarchy, timing and precedent status. |
| <b>Hybrid retrieval</b>                     | Combining exact keyword search with meaning-based (vector) search, so that both precise legal terms and loosely worded questions can be matched to the right documents.                                             |
| <b>BM25</b>                                 | A well-established keyword-search ranking method, useful for finding exact terms such as case numbers or section numbers.                                                                                           |
| <b>Vector search / embeddings</b>           | A method of representing text as numbers (“embeddings”) so that a computer can find documents with similar meaning, even if they use different words.                                                               |
| <b>OCR (Optical Character Recognition)</b>  | Software that converts scanned images of text into machine-readable text.                                                                                                                                           |
| <b>Metadata</b>                             | Structured information about a document — such as its date, court, status or citations — kept separately from the document's main text.                                                                             |
| <b>CEPEJ</b>                                | The Council of Europe's European Commission for the Efficiency of Justice, which publishes ethical guidance on AI in judicial systems.                                                                              |
| <b>NIST AI RMF</b>                          | The U.S. National Institute of Standards and Technology's AI Risk Management Framework, a general model for governing AI risk.                                                                                      |
| <b>OWASP</b>                                | A nonprofit organization known for publishing widely used security guidance, including a “Top 10” list of risks specific to LLM-based applications.                                                                 |